



Introduction to Starburst

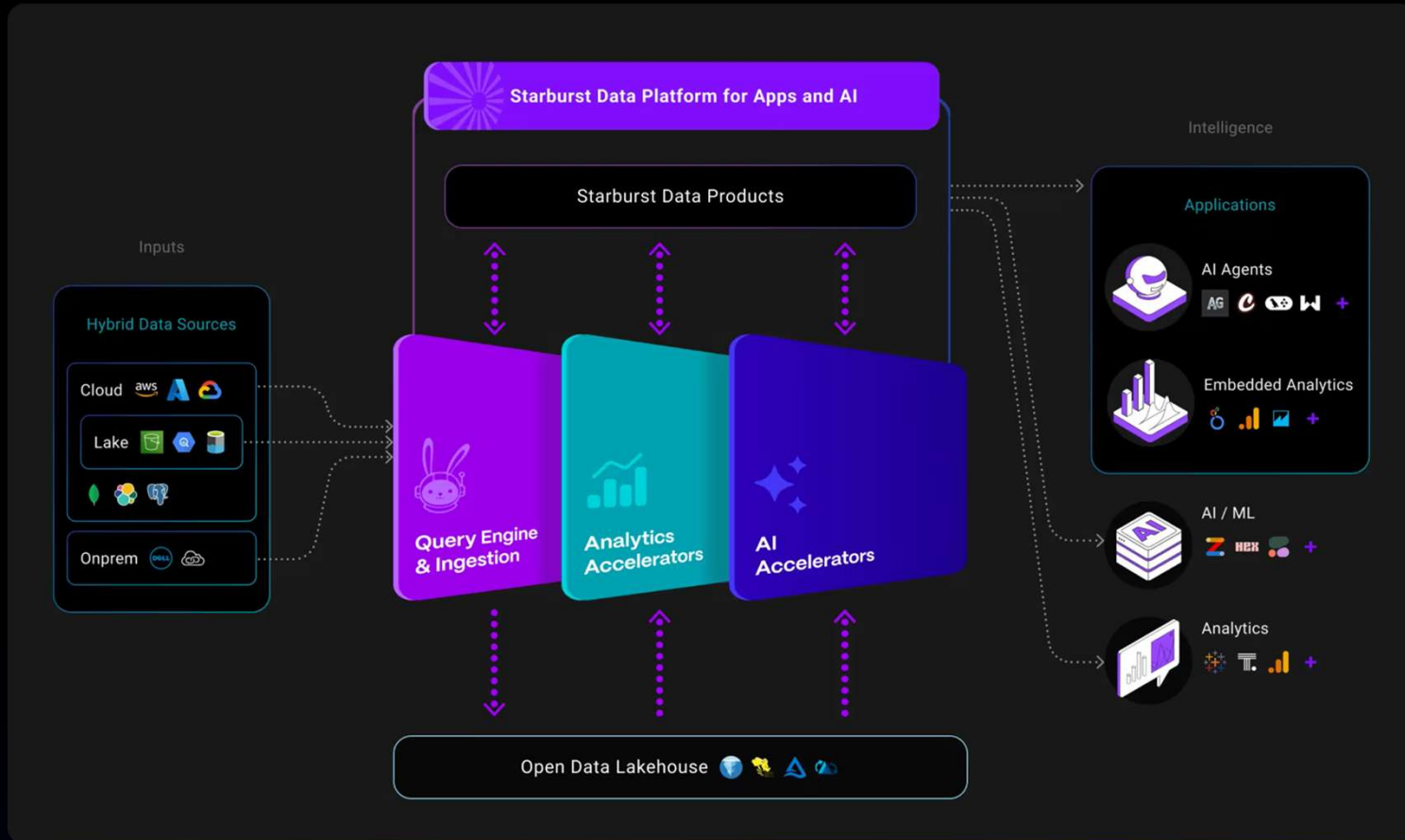
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Lesson objectives

Introduction to Starburst

1. Explain what Starburst is and how analysts use it.
2. Outline the key benefits of using Starburst.
3. Describe how Starburst fits into the modern data ecosystem.
4. Navigate the Starburst Enterprise web UI.

What is Starburst?





- Built on Trino – MPP query engine
- Two platforms – Starburst Enterprise or Starburst Galaxy
- Query data where it lives (no movement)
- Connect to all your data and your BI tools
- Robust data governance – Security and data discovery
- AI functionality

Starburst in the Modern Data Ecosystem

Data Access Layer Across the Ecosystem

- **Challenge today:** Data lives in multiple places — data lakes (S3, ADLS, GCS), data warehouses (Snowflake, BigQuery, Redshift), databases, SaaS apps, etc.
- **Starburst's role:** Acts as a federated query engine so you can query across all those systems with a single SQL interface, without moving or duplicating data.

Foundation for Data Mesh & Data Products

- **Modern trend:** Many organizations are moving to data mesh or domain-driven ownership.
- **Starburst's role:** Provides the tooling to create and publish data products — curated, governed, discoverable datasets that analysts and data scientists can use across domains.

Starburst in the Modern Data Ecosystem

Empowering Different Personas

- **Data Consumers** (analysts, business users): Query all data through BI tools or directly in SQL.
- **Data Engineers**: Build pipelines more efficiently by eliminating data movement.
- **Data Scientists/AI teams**: Use trusted data products as input for ML/AI models.
- **Platform Administrators**: Govern, secure, and optimize data access centrally.

Governance, Security, and Compliance

- **Challenge**: Companies need to secure data consistently across many systems.
- **Starburst's role**: Centralizes access control, security policies, lineage, and auditing across data sources — reducing risk and ensuring compliance.

Starburst in the Modern Data Ecosystem

Performance & Cost Optimization

- **Challenge:** Warehouses are powerful but expensive; lakes are cheap but slow.
- **Starburst's role:** Optimizes query execution across sources, supports caching, and lets teams choose the right engine for the right workload — lowering cost while maintaining speed.

AI & Advanced Analytics

- **Challenge:** AI needs high-quality, governed, and connected data.
- **Starburst's role:** Provides AI teams with one governed access layer to all enterprise data, making training data pipelines faster and more trustworthy.

How Do Analysts Use Starburst?

Unified Data Access

Problem: Data is scattered across warehouses, lakes, and databases.

Starburst Benefit: Analysts query all those sources from one place without needing to move or copy data.

Example: Run a single SQL query joining customer data in Snowflake with product usage logs in S3.

Querying With SQL

Analysts use familiar SQL in Starburst's web UI (or connected BI tools).

They can:

- Filter, aggregate, and join tables across sources.
- Run ad-hoc analysis for business questions.
- Save queries and share with teammates.

Using Data Products

Analysts prefer curated, trusted datasets instead of stitching raw tables together.

Data products in Starburst provide:

- Clean, documented views (e.g., Customer 360, Sales Performance).
- Ownership and quality indicators.

This helps analysts focus on analysis instead of wrangling data.

Exploratory & Ad-Hoc Analysis

Analysts answer business questions like:

- "Which customers are at risk of churn?"
- "Which campaigns drove the most revenue last quarter?"

They can query directly in Starburst or use it as the backend for BI tools like Tableau, Power BI, Looker.

How Do Analysts Use Starburst? (cont'd)

Performance & Scale

Analysts run queries on **big data** without worrying about infrastructure.

Starburst optimizes query execution across sources, so analysts focus on insights rather than performance tuning.

AI-Powered Assistance

If enabled, analysts can use **natural language queries**:

- *“Show me sales growth by region over the past 3 months.”*

Starburst (with AI features) can translate plain English into SQL and return results.

In short: Data analysts use Starburst as their **single SQL-powered gateway** to all company data, with the bonus of curated data products and AI assistance that help them deliver faster, trusted insights.

Hands-On Exercise

Access your lab
environment (5 minutes)

Lesson summary

Introduction to Starburst

- Starburst is a platform built on Trino, an MPP query engine.
- It allows users to query data where it lives without moving or duplicating it.
- It connects to all data sources and BI tools, offering robust data governance (security, discovery) and AI functionality.
- It acts as a federated query engine, providing a single SQL interface to query across diverse systems like data lakes, data warehouses, and SaaS apps.
- It forms a foundation for Data Mesh and Data Products, enabling the creation of curated, governed, and discoverable datasets.
- Starburst empowers various users (analysts, engineers, data scientists, administrators) by centralizing data access, eliminating data movement, and providing trusted data for ML/AI.
- It centralizes governance, security, and compliance by managing access control, policies, and auditing across sources.

Lesson summary (cont'd)

Introduction to Starburst

- By using Starburst, analysts gain unified data access, querying scattered data from one place using familiar SQL in Starburst's web UI or connected BI tools.
- They utilize data products—clean, documented views—to focus on analysis rather than data wrangling.
- They perform exploratory and ad-hoc analysis to answer business questions, directly or via BI tools.
- Starburst handles performance and scale for big data queries, letting analysts focus on insights.